

Campfire Forest Night: A Dynamic 3D Nighttime Scene using OpenGL

Project Proposal

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1 Project Description

This project implements a real-time, interactive 3D nighttime forest campsite rendered entirely using Python and the PyOpenGL fixed-function pipeline. The scene depicts a moonlit clearing surrounded by dense coniferous trees, centered on a crackling campfire with a pitched tent nearby. An animated owl perches on a tree branch, turning its head and flapping its wings. The entire scene is procedurally generated — all geometry, textures, and particle effects are created algorithmically in code, requiring no external 3D models or image files. The project demonstrates core computer graphics concepts including hierarchical transformations (`glPushMatrix/glPopMatrix`), positional and directional lighting (`GL_LIGHT0`, `GL_LIGHT1`), material emission, real-time animation, procedural texture generation, and an interactive orbit camera with full mouse and keyboard control. Frame-rate-independent animation ensures smooth motion regardless of hardware performance.

2 Objects in the Scene

- ▶ Undulating forest terrain (44×44 mesh)
- ▶ Moon with emissive glow
- ▶ 6 detailed foreground trees with bark texture
- ▶ 90 procedural stars in the night sky
- ▶ 15 distant silhouette trees
- ▶ 13 uneven stones forming a fire ring
- ▶ Layered cone foliage with wind sway
- ▶ 5 crossed bark-textured fire logs
- ▶ Root flares and branches per tree
- ▶ 17 glowing ember-bed spheres
- ▶ Hierarchical owl (body, head, wings, eyes, beak, talons)
- ▶ 5 additive flame layers with pulsing scale
- ▶ Canvas-textured tent with entrance & flaps
- ▶ 14 translucent smoke particles
- ▶ Tent ridge pole, entrance pole & guy ropes
- ▶ 18 rising ember particles
- ▶ 52 procedural grass tufts
- ▶ 10 fireflies on elliptical orbits
- ▶ 19 forest floor rocks
- ▶ 8 soft contact shadows

3 Extra Features

- ★ Three procedural textures generated in code (ground, bark, canvas) — no image files needed
- ★ Flickering positional firelight with physical attenuation (constant + linear + quadratic)
- ★ EXP2 atmospheric depth fog with real-time toggle (press F)
- ★ Interactive orbit camera — mouse drag, arrow keys, zoom, elevation, and reset controls
- ★ Additive-blended fire glow, flames, and embers for realistic campfire lighting
- ★ Deterministic seeded randomness — scene renders identically on every run
- ★ Pause/resume animation (Spacebar) for inspection of any frame
- ★ 7+ independent animation systems running simultaneously with frame-rate independence